

# A Classical Permanent Inequality Answers Problems 1 and 2 of arXiv:2306.14807

Literature-implied answer note

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**Status.** `literature_implied_answer` (full answers to Problems 1 and 2). The relation to the source questions is identified here; Marcus and Paksoy did not claim to answer arXiv:2306.14807.

## Source questions

Garcia, O’Loughlin, and Yu ask on page 18, Section 10, Problem 1 of [1] whether, for  $x_1, \dots, x_n \in \mathcal{H}$ ,

$$\frac{1}{\sqrt{n!}} \prod_{i=1}^n \|x_i\| \leq \|x_1 \odot \dots \odot x_n\|.$$

Their Problem 2 asks whether, for  $A_1, \dots, A_n \in \mathcal{B}(\mathcal{H})$ ,

$$\frac{1}{\sqrt{n!}} \sup_{\|x\|=1} \prod_{i=1}^n \|A_i x\| \leq \|A_1 \odot \dots \odot A_n\|.$$

## Supporting literature

Marcus’s permanent analogue of Hadamard’s theorem states that every Hermitian matrix  $G = (g_{ij})_{i,j=1}^n$  that is positive semidefinite satisfies

$$\text{per } G \geq \prod_{i=1}^n g_{ii}. \tag{1}$$

This is Corollary (3) on page 494 of [2]. An accessible modern supporting source is Paksoy [3]: Theorem 2.4 on page 73 gives the Gram-matrix/symmetric-tensor identity, and formula (3.15) on page 76 records (1).

## Identification

**Proposition.** *Problems 1 and 2 of [1] have affirmative answers. The constant in Problem 1 is sharp whenever  $\dim \mathcal{H} \geq n$ .*

*Proof.* Let  $u = x_1 \otimes \dots \otimes x_n$  and let

$$\mathcal{S}_n = \frac{1}{n!} \sum_{\pi \in S_n} \hat{\pi}$$

be the orthogonal symmetrization projection used in [1]. For the Gram matrix  $G = (\langle x_i, x_j \rangle)$ , self-adjointness and idempotence of  $\mathcal{S}_n$  give

$$\begin{aligned} \|x_1 \odot \cdots \odot x_n\|^2 &= \langle \mathcal{S}_n u, \mathcal{S}_n u \rangle = \langle u, \mathcal{S}_n u \rangle \\ &= \frac{1}{n!} \sum_{\pi \in S_n} \prod_{i=1}^n \langle x_i, x_{\pi(i)} \rangle = \frac{1}{n!} \text{per } G. \end{aligned}$$

Since  $G$  is Hermitian positive semidefinite, (1) yields

$$\|x_1 \odot \cdots \odot x_n\|^2 \geq \frac{1}{n!} \prod_{i=1}^n \|x_i\|^2.$$

Taking square roots proves Problem 1. If  $x_1, \dots, x_n$  are mutually orthogonal and nonzero,  $G$  is diagonal and equality holds, proving sharpness.

Now let  $x \in \mathcal{H}$  be a unit vector. The definition of the symmetric product of operators in [1] gives

$$(A_1 \odot \cdots \odot A_n)x^{\otimes n} = (A_1 x) \odot \cdots \odot (A_n x).$$

Because  $\|x^{\otimes n}\| = 1$ , Problem 1 applied to  $A_1 x, \dots, A_n x$  gives

$$\|A_1 \odot \cdots \odot A_n\| \geq \frac{1}{\sqrt{n!}} \prod_{i=1}^n \|A_i x\|.$$

Taking the supremum over unit vectors  $x$  proves Problem 2. □

## Scope and search evidence

The source’s exact arXiv id, title phrases, “symmetric tensor permanent,” and “permanent positive semidefinite product diagonal” were checked in the run’s four lightweight indexes and in primary-source web search. No prior run packet was found. The decisive theorem is classical, so this is not a new result. Problems 3-9 of [1], concerning sharp operator lower bounds and spectra, remain outside this identification.

## Review recommendation

Verify the source normalization  $\mathcal{S}_n = (n!)^{-1} \sum \hat{\pi}$  and the one-line passage from the vector identity to the operator action on  $x^{\otimes n}$ . The permanent theorem itself is explicitly recorded in both supporting sources.

## References

- [1] S. R. Garcia, R. O’Loughlin, and J. Yu, *Symmetric and Antisymmetric Tensor Products for the Function-Theoretic Operator Theorist*, arXiv:2306.14807, 2023.
- [2] M. Marcus, *The permanent analogue of the Hadamard determinant theorem*, Bull. Amer. Math. Soc. 69 (1963), 494–496, doi:10.1090/S0002-9904-1963-10975-1.
- [3] V. E. Paksoy, *A Permanent Inequality for Positive Semidefinite Matrices*, Electron. J. Linear Algebra 39 (2023), 71–77, doi:10.13001/ela.2023.7701.